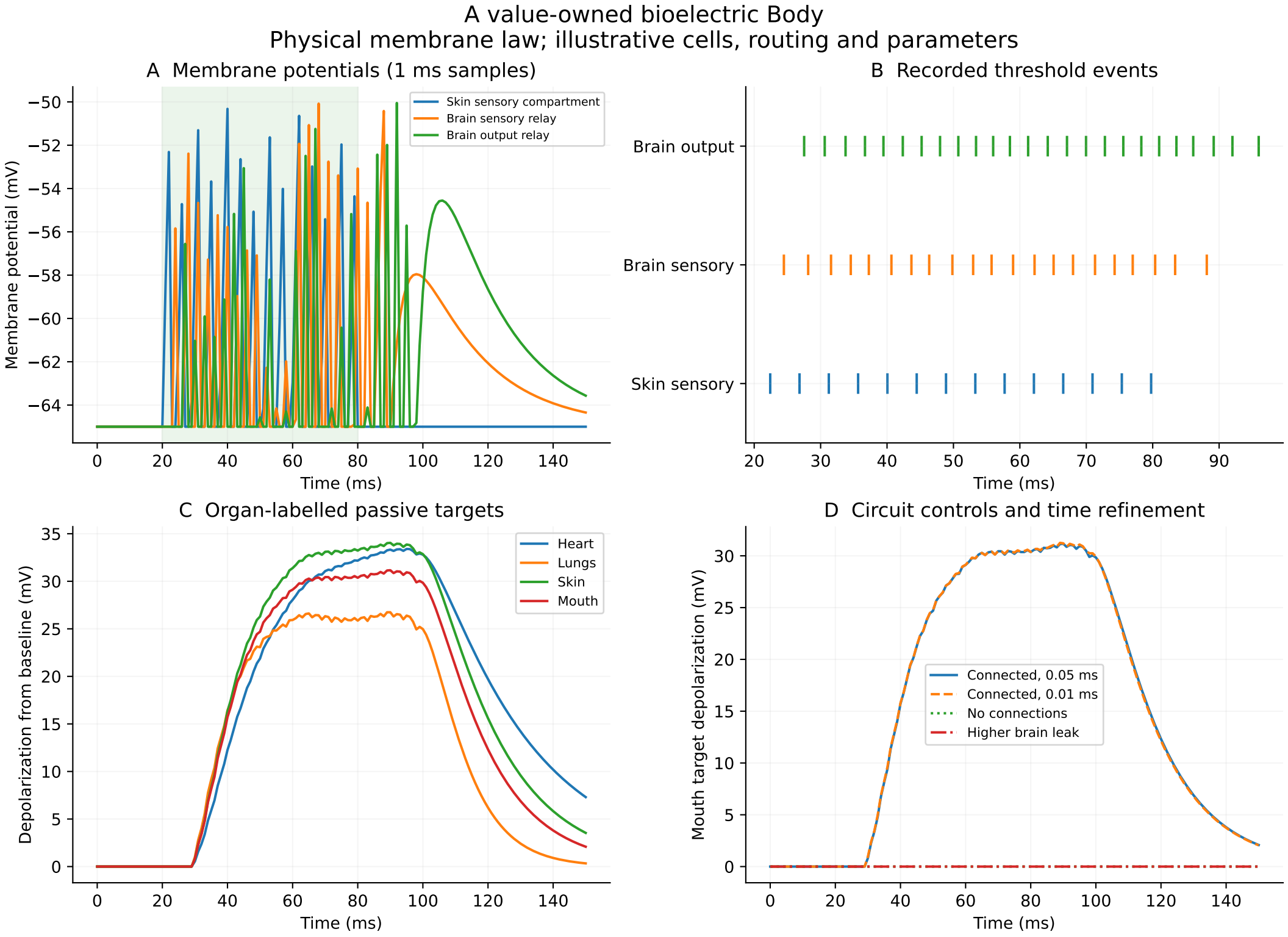
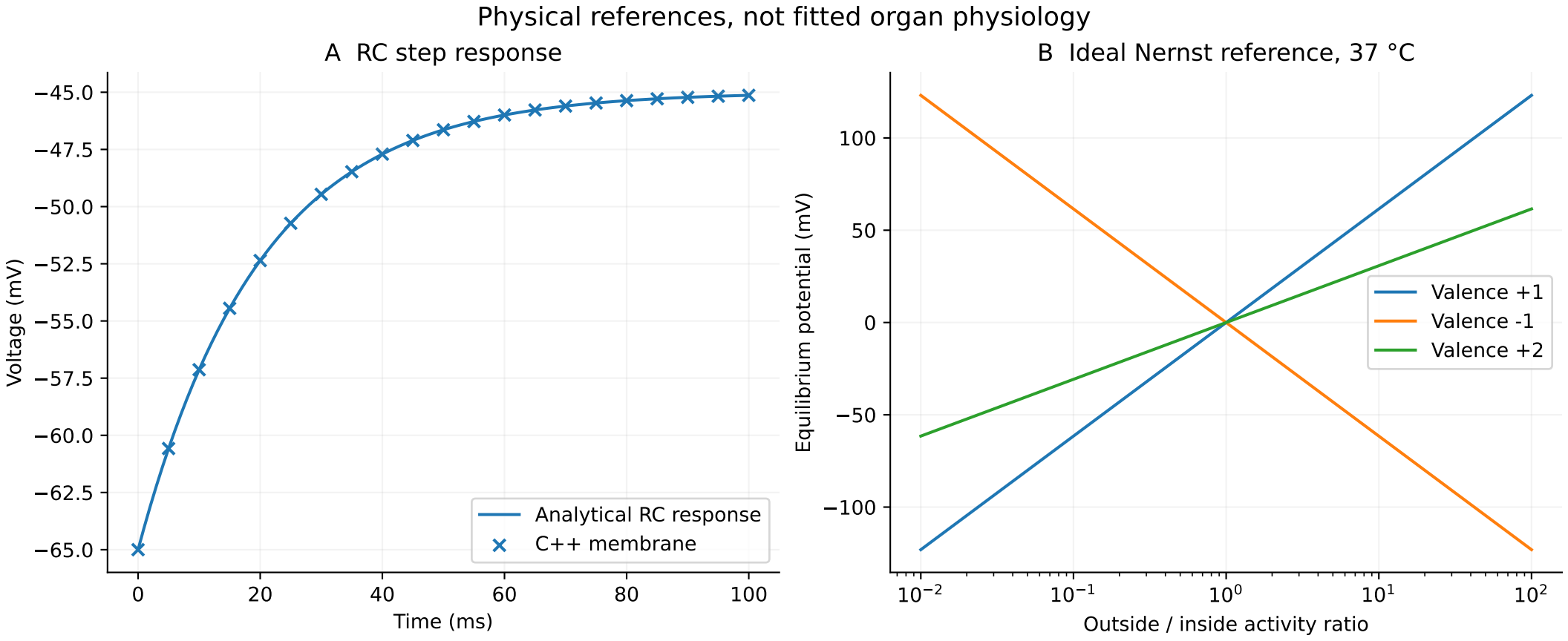

A value-owned bioelectric Body
Physical membrane law; illustrative cells, routing and parameters
A Membrane potentials (1 ms samples)
Membrane potential (mV)
Time (ms)
Skin sensory compartment
Brain sensory relay
Brain output relay
B Recorded threshold events
Brain output
Brain sensory
Skin sensory
Time (ms)
C Organ-labelled passive targets
Depolarization from baseline (mV)
Time (ms)
Heart
Lungs
Skin
Mouth
D Circuit controls and time refinement
Mouth target depolarization (mV)
Time (ms)
Connected, 0.05 ms
Connected, 0.01 ms
No connections
Higher brain leak


Physical references, not fitted organ physiology
A RC step response
Voltage (mV)
−45.0
−47.5
−50.0
−52.5
−55.0
−57.5
−60.0
−62.5
−65.0
0
20
40
60
80
100
Time (ms)
Analytical RC response
C++ membrane
B Ideal Nernst reference, 37 °C
Equilibrium potential (mV)
100
50
0
−50
−100
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Outside / inside activity ratio
Valence +1
Valence -1
Valence +2